Monte Carlo simulation of stochastically heated grains and comparison with the Guhathakurta–Draine method

Abstract

We implement the Monte Carlo (MC) algorithm of Draine & Anderson (1985) for tracking the temperature of a single interstellar dust grain through individual photon-absorption events, extend it with a selectable heat-capacity model, and reproduce Figures 24.5 and 24.6 of Draine (2011). The standalone implementation is kept for visualization; a subroutine-form API is then used to integrate $P(T)$ over a grain size distribution and build an emergent SED. The SED builder exposes a runtime flag `use_mc_for_large_grain`. In the default mode (`= F`), small (stochastic) grains use the Monte Carlo engine while large (equilibrium) grains use the analytic `calc_Teq` solver – the same hybrid logic the production solver uses, but with MC replacing the Guhathakurta & Draine (1989) matrix solver in the stochastic branch. The implementation divides the radiation field into discrete photon absorptions and continuous heating at an adaptive wavelength cutoff $\lambda_c(a) = hc/[\delta T_{\rm thresh} T_{\rm eq} C_{\rm tot}(T_{\rm eq})]$, limits each cooling step to a 5% temperature change, integrates every sampled waiting interval completely, and stops with an explicit status if a step or sample budget is exhausted. Each run first relaxes the grain from its radiative-equilibrium temperature for a fixed preparation time that is excluded from all recorded quantities. Neutral and ionized PAHs are calculated separately, and each grain spectrum is normalized to its absorbed power. For the HD23 astrodust+PAH model in the Mathis field at $\log U = 0.20$, the default-mode MC SED agrees with the production solver to within 0.2% in the band-integrated 5–30, 30–300, and 300–3000 μm power (ratios 0.998, 1.001, 1.001) and is 3.9% low in 1–5 μm (ratio 0.961). The all-MC mode gives the same band ratios to 0.0003, and the radiation-equivalent MC temperature of every equilibrium-gate grain lies within –0.13% and +0.01% of $T_{\rm eq}$ for $a = 0.032$–5 μm. Before the spectral normalization the recorded trajectories close their energy budget to between −1.08% and +0.08% of the absorbed energy, grain by grain. The photon-wavelength sampler follows the supported absorption spectrum: five graphite-grain tests of 2×10^6 draws each found no photons below the 0.0912 μm Lyman edge, and the measured mean energies agree with the exact means of the implemented distributions within 0.60 standard errors (Section 4.3).

Note on the Mathis ISRF. The `radfield.f90::J_Mathis` default uses the corrected 4000 K dilution factor 1.65×10^{-13} and a 2.725 K CMB, and every MC calculation reported here uses it. The selectable `use_mathis_corrected = .false.` setting (namelist `radiation_field='mathis_orig'` of the SED and comparison drivers) reproduces the original 10^{-13} factor and a 2.9 K CMB.

Measurements quoted here. The values were measured on 2026-10-31 from the Mathis-field calculations in `SEDust_JKAS/results/20261031_mc_photon_support`, using the current `sed/output` reference tables in each branch. The result directory retains the input and source hashes, run logs, recorded energy budgets, and full temperature distributions. Separate single-grain PAH and large-grain calculations use the same source and optical data and are retained there under `report_extra`. The calculations used gfortran 16.2 (`-O2`) on a machine with two 18-core Intel Xeon Gold 6154 processors (72 hardware threads).

Contents

1. Introduction **2**

2. Algorithm **3**

2.1 Two-regime treatment of the absorbed radiation field 3

2.2 Cooling ODE between events 4

2.3 The Monte Carlo loop 4

2.4 Integrator for the cooling ODE 5

3. Implementation **6**

3.1 Module layout 6

3.2 Stationary recording and transient experiments 7

3.3 Energy bookkeeping 8

4. Validation against PIIM Figures 24.5 and 24.6 **9**

4.1 Figure 24.5: $T(t)$ trajectories 9

4.2 Figure 24.6: $dP/d\ln T$ 9

4.3 Photon wavelength distribution 9

5. SED builder over a grain size distribution **13**

5.1 SED accumulation 13

5.2 Energy normalization and charge-resolved PAH 13

5.3 OpenMP parallelism 14

6. Comparison with the Guhathakurta–Draine matrix solver **14**

6.1 Two solver modes 15

6.2 Equilibrium grains in the all-MC mode 15

6.3 When to use each mode 15

6.4 Step budget 16

6.5 Single-grain $P(T)$ cross-check 16

7. Adaptive-grid MC engines **21**

7.1 Adaptive DA85 cutoff $\lambda_c(a)$ 22

1. Introduction

A dust grain in the interstellar medium absorbs starlight photons discretely. When the grain is macroscopic ($a \gtrsim 0.03\,\mu$m), many absorption events occur per radiative cooling time and the grain sits at a well-defined equilibrium temperature $T_{\rm eq}(a)$ at which the mean absorbed power balances the radiative cooling. Smaller grains, however, do not have enough heat capacity to absorb the energy of a single starlight photon without making a large, transient excursion in temperature.

Each photon absorption raises the grain's temperature by tens to hundreds of Kelvin; between events the grain cools radiatively back toward $T_{\rm eq}$. The temperature history $T(t)$ is a sequence of sharp upward spikes followed by long, slow cooling tails, as illustrated in Figure 24.5 of Draine (2011).

Because the emission spectrum scales steeply with temperature ($\propto T^4$ in total luminosity, with strong wavelength dependence through the Planck function and the cross section), the spectral energy distribution (SED) of small grains is dominated by the high-temperature spikes, not by $T_{\rm eq}$. Quantitative treatment requires the probability distribution $P(T)$, which is the fraction of time the grain spends at each temperature in steady state.

There are two standard ways to obtain $P(T)$:

1. **Monte Carlo** (Draine & Anderson, 1985). Simulate the stochastic sequence of absorption events directly and accumulate a time-weighted histogram of $T(t)$.
2. **Discretized steady-state matrix solve** (Guhathakurta & Draine, 1989; Draine & Li, 2001). Discretize T into bins, build a transition matrix from absorption (upward) and Planck-averaged emission (downward) rates, and solve for the steady state.

The matrix method is more efficient and is what the production astrodust solver uses (`p_sub.f90::calc_P`). The MC method is conceptually simpler, has fewer numerical knobs, and provides a useful independent cross-check. This document records the MC implementation kept under `SEDust/mc/` and its validation.

Document layout. Section 2. derives the MC algorithm and the cooling ODE used between events. Section 3. describes the Fortran module structure, the selectable heat-capacity model, the stationary preparation, and the energy bookkeeping. Section 4. presents the reproduction of Draine (2011) Figures 24.5 and 24.6. Section 5. describes the subroutine-form API and the SED builder that integrates $P(T)$ over a grain size distribution. Section 6. compares the MC-based SED and single-grain $P(T)$ with the Guhathakurta–Draine matrix solver on identical inputs, and Section 7. describes the histogram engines and the adaptive cutoff.

2. Algorithm

2.1 Two-regime treatment of the absorbed radiation field

Consider a spherical grain of radius a in an isotropic radiation field of mean intensity $J_\lambda(\lambda)$. The radiative energy density per unit wavelength is

$$u_\lambda = \frac{4\pi}{c} J_\lambda \quad [\mathrm{erg\,cm^{-3}}\ \mu\mathrm{m}^{-1}]. \tag{1}$$

The grain absorbs photons of wavelength λ at a rate per unit wavelength

$$\frac{dN_{\rm abs}}{dt\,d\lambda} = \pi a^2\, Q_{\rm abs}(a,\lambda)\, \frac{c\,u_\lambda\,\lambda}{hc} = \frac{\pi a^2}{h}\, Q_{\rm abs}(a,\lambda)\, u_\lambda\,\lambda, \tag{2}$$

where the photon energy is $E = hc/\lambda$ and we have used $u_\lambda/E = (u_\lambda\,\lambda)/(hc)$ as the photon number density per unit wavelength, multiplied by c to get a photon flux density and by $\pi a^2\, Q_{\rm abs}$ to get the absorption rate.

Draine & Anderson (1985) split this rate into a continuous part (long wavelengths, where photons arrive so frequently that the grain heat content increases smoothly) and a stochastic part (short wavelengths, where individual events are

resolved). Pick a cutoff λ_c: the standalone driver `main_mc.x` uses the namelist value `lam_c_um` (default 1000 μm), while the SED builder and the single-grain comparison driver use the adaptive cutoff of Section 7.1. The continuous-heating rate per unit surface area is

$$H(a) \equiv \frac{c}{4}\int_{\lambda_c}^{\infty} Q_{\rm abs}(a,\lambda)\,u_\lambda\,d\lambda \quad [\mathrm{erg\,cm^{-2}\,s^{-1}}]. \tag{3}$$

The stochastic event rate is

$$\Gamma_{\rm evt}(a) \equiv \int_0^{\lambda_c} \frac{dN_{\rm abs}}{dt\,d\lambda}\,d\lambda \quad [\mathrm{s^{-1}}]. \tag{4}$$

The division depends on λ_c and the radiation field, and both terms are retained in the calculation. For the smallest astrodust grains in the Mathis field at $U = 1.585$, the adaptive cutoff reaches the long-wavelength grid limit ($3.936\times10^4\,\mu$m), so $H(a)$ contains only the CMB absorption longward of that wavelength and is of order 10^{-13} of the event power (Section 7.1).

The tabulated wavelength grid generally has one interval straddling λ_c. That interval is split at λ_c by linear interpolation of the integrand: its blueward part is added to $\Gamma_{\rm evt}$ (and to the photon-sampling CDF) and its redward part to $H(a)$, so the cutoff bin is dropped from neither term. The sampling CDF is built from the same split integral as $\Gamma_{\rm evt}$ and has its last node pinned at λ_c, so a sampled photon always satisfies $\lambda \le \lambda_c$. A grain with no absorption blueward of λ_c has $\Gamma_{\rm evt} = 0$ and belongs on the equilibrium path; `grain_setup_from_cabs` stops with a diagnostic rather than divide by a zero event rate.

2.2 Cooling ODE between events

Let $E_{\rm grain}(T)$ be the total grain enthalpy and $C_{\rm tot}(T) \equiv dE_{\rm grain}/dT$ its total heat capacity. Between absorption events, the grain's energy evolves according to

$$\frac{dE_{\rm grain}}{dt} = P_{\rm heat} - P_{\rm emit} = 4\pi a^2\left[H(a) - \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4\right], \tag{5}$$

where $\langle Q_{\rm abs}\rangle_T$ is the Planck-averaged absorption efficiency,

$$\langle Q_{\rm abs}\rangle_T \equiv \frac{\int Q_{\rm abs}(a,\lambda) B_\lambda(T)\,d\lambda}{\int B_\lambda(T)\,d\lambda}. \tag{6}$$

Dividing through by $C_{\rm tot}$ gives the cooling ODE used in the MC:

$$\boxed{\frac{dT}{dt} = \frac{4\pi a^2}{C_{\rm tot}(T)}\left[H(a) - \langle Q_{\rm abs}\rangle_T\,\sigma_{\rm SB}\,T^4\right].} \tag{7}$$

Equation (7) is identical to Eq. (2.9) of Draine & Anderson (1985), which writes it in the equivalent form $dT/dt = (3/(aC_{\rm vol}(T)))(H - \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4)$ using the heat capacity per unit volume $C_{\rm vol} = C_{\rm tot}/(4\pi a^3/3)$. We use the total form because the enthalpy routines (`sed/src/enthalpy_v2.f90::enthalpy_DL01` and the astrodust enthalpy module) return the total grain enthalpy in erg. The factor $4\pi a^2$ in our form is the grain *surface* area (heating and emission are surface effects), and the $1/C_{\rm tot}$ absorbs the grain volume implicit in the total heat capacity. The engine tabulates $\langle Q_{\rm abs}\rangle_T$ on a 2000-point logarithmic grid from 1 to 10^4 K and uses $\langle Q_{\rm abs}\rangle_{1\,\rm K}$ below 1 K.

2.3 The Monte Carlo loop

Starting from (t_0, T_0), one step of the MC simulation does the following.

1. **Sample the inter-arrival time.** The event-arrival process is Poissonian with rate $\Gamma_{\rm evt}$, so the waiting time $\Delta t = -\ln U_1/\Gamma_{\rm evt}$ where $U_1 \in (0,1]$ is a uniform random number.

2. **Integrate Eq. (7) over $[t_0, t_0+\Delta t]$.** The grain cools (or, in the rare regime where $H > \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$, heats) according to the continuous part of the radiation field. During this step we accumulate the time-weighted contribution to the $P(T)$ histogram and to the integrated emitted energy.

3. **Sample the absorbed photon.** Choose a wavelength λ from the normalized PDF $p(\lambda) \propto dN_{\rm abs}/(dt\, d\lambda)$ on the supported interval below λ_c. The cumulative probability is stored on the physical wavelength knots. A binary search selects a positive-probability interval, and the integral of its linear photon density is inverted within that interval.

4. **Apply the enthalpy jump.** The absorbed photon energy $E_{\rm phot} = hc/\lambda$ goes into internal vibrational energy. Solve

$$E_{\rm grain}(T_{\rm after}) = E_{\rm grain}(T_{\rm before}) + E_{\rm phot} \tag{8}$$

   for $T_{\rm after}$. The enthalpy table locates a bracket; a safeguarded Newton–bisection inversion then evaluates the analytic $E_{\rm grain}(T)$ within that bracket. When adjacent representable temperatures bracket the target energy, the closer endpoint is accepted within a bounded enthalpy error.

The loop is repeated for a prescribed number $N_{\rm evt}$ of recorded events, or until a prescribed recorded time $t_{\rm max}$ is reached. The calculations in this report use $N_{\rm evt} = 5\times10^4$ (SED default), 2×10^5 and 5×10^5 events per grain.

2.4 Integrator for the cooling ODE

For a 100 Å graphite grain in the $U = 1$ Mathis field the measured event rate is $\Gamma_{\rm evt} = 2.06\times10^{-3}\,{\rm s}^{-1}$, a mean inter-arrival time of 486 s, which is long compared with the radiative cooling after a photon spike. The cooling ODE (7) is therefore mildly stiff: rapid initial decay followed by a long, near-equilibrium tail. Explicit Euler is prohibitively slow because the step size near equilibrium is set by the restoring rate, which is large.

The right-hand side of Eq. (7) is dominated, after the photon spike, by the T^4 Stefan–Boltzmann emission term: with $H(a) \ll \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$ the equation reduces to $dT/dt \simeq -(4\pi a^2/C_{\rm tot})\langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4$. Both $C_{\rm tot}(T)$ and $\langle Q_{\rm abs}\rangle_T$ depend on T, so the trajectory is not a single exponential – it falls steeply at high T and slowly near $T_{\rm eq}$. Any closed-form fit (exponential, power-law T^{-3} from constant-C Stefan–Boltzmann) is an approximation; we therefore integrate Eq. (7) numerically and, when a continuous $T(t)$ is needed for a figure, dump the actual integrator sub-step samples rather than reconstructing from event boundaries.

We use a linearly-implicit (exponential) step. At the current temperature T_0, we form a one-sided finite difference to estimate the local slope

$$\Lambda(T_0) \;\equiv\; \left.\frac{d(dT/dt)}{dT}\right|_{T_0} \approx \frac{(dT/dt)_{1.01\,T_0} - (dT/dt)_{T_0}}{0.01\,T_0}. \tag{9}$$

When $\Lambda < 0$ the dynamics is locally a stable relaxation toward a local equilibrium temperature $T_{\rm eq,loc} = T_0 - (dT/dt)_{T_0}/\Lambda$, and the linearized solution

$$T(t_0+dt) \;=\; T_{\rm eq,loc} + \left(T_0 - T_{\rm eq,loc}\right) e^{\Lambda\, dt} \tag{10}$$

is exact for the linearized equation. We step by

$$dt = \min\left(dt_{\rm left},\ \frac{\ln 2}{|\Lambda|},\ f_{\rm cap}\,\frac{T_0}{|dT/dt|_{T_0}}\right), \qquad f_{\rm cap} = 0.05; \tag{11}$$

once $|T_0 - T_{\rm eq,loc}|/T_0 < 10^{-3}$ we take the full remaining $dt_{\rm left}$ in a single step at $T_{\rm eq,loc}$. The cap $\ln 2/|\Lambda|$ halves the distance to local equilibrium per step. The cap on the *fractional* temperature change, $f_{\rm cap} = 5\%$ per step, bounds the linearization error: the linearized solution (10) is exact only for a locally *linear* ODE, whereas the true cooling is strongly super-linear ($dT/dt \propto -\langle Q_{\rm abs}\rangle_T T^4$), so over a long step the linearization lags the faster early cooling and the grain lingers at high T. With the cap in place the recorded trajectories still emit slightly more than they absorb (Section 3.3); this residual does not change between 5×10^4 and 5×10^5 events and is a property of the step, not of the sampling. The engines continue cooling through each sampled waiting interval, in chunks of internal steps (300 for the adaptive engines, 4000 for the fixed-grid engine). A budget of 10^6 steps applies to each waiting interval by default; exceeding it returns a failure status and the unintegrated time rather than a truncated trajectory.

The $P(T)$ histogram and the emitted-energy bookkeeping sub-sample each step at $N_{\rm sub} = 8$ interior points of the sub-step trajectory, each carrying weight $dt/N_{\rm sub}$, so a step that spans several histogram bins deposits its time weight across all of them. For $\Lambda \geq 0$ the step is a forward Euler step, also subject to the $f_{\rm cap}$ limit.

The engine imposes a numerical temperature floor of 0.5 K. A grain at the floor whose cooling rate still points downward stays there for the rest of its waiting interval, and that residence is reported separately (`floor_time_out`). The floor is a lower limit of the numerical representation, not a physical temperature: below 1 K the cooling uses $\langle Q_{\rm abs}\rangle_{1\,\rm K}$, and the formal balance temperatures of the smallest grains lie far below 0.5 K (Section 7.1).

3. Implementation

3.1 Module layout

The implementation lives under `SEDust/mc/` and is split into small modules that each do one thing. Throughout, cgs units are used (erg, cm, s) except that wavelength is in μm to match the rest of the astrodust code.

- `mc_heatcap.f90`: enthalpy $E_{\rm grain}(T, a)$ and heat capacity $C_{\rm tot}(T) = dE_{\rm grain}/dT$ for selectable compositions (Table 1). $C_{\rm tot}$ is computed by central differences on $E_{\rm grain}$, which works for all heat-capacity models without composition-specific analytic derivatives.
- `mc_rng.f90`: thread-local pseudo-random number generator (xorshift64*, period $2^{64} - 1$) wrapped in a derived type `rng_t`. Each OpenMP thread owns its own state, so no locking is needed inside the MC inner loop.
- `mc_grain_type.f90`: derived type `mc_grain_t` holding all single-grain pre-tabulated state (λ grid, $Q_{\rm abs}$, u_λ, $H_{\rm cont}$, $\Gamma_{\rm evt}$, photon-wavelength CDF, T grid, $E_{\rm grain}(T)$, $\langle Q_{\rm abs}\rangle_T$). Instances are local to a thread.
- `mc_engine.f90`: the engine. Two setup entry points: `grain_setup_graphite` (D16 graphite sphere Q on a logarithmic grid from 10^{-3} to 10^4 μm, used by the standalone validation path) and `grain_setup_from_cabs` (arbitrary $C_{\rm abs}(\lambda)$ on a caller-supplied grid, used by the SED builder and the single-grain comparison driver). Three MC drivers with the same signature shape, differing only in the histogram strategy: `mc_run_engine` (fixed

logarithmic grid on [0.5, 5000] K), `mc_run_engine_2pass`, and `mc_run_engine_buffered`. All three return the time-weighted $P(T)$ histogram (as bin edges $T_{\rm edge}$, dP/dT, and $dP/d\ln T$) and the integrated absorbed and emitted energies; the adaptive engines additionally return the selected grid type (`is_log_out`) and the observed $T_{\rm min}/T_{\rm max}$. Each driver reports completion status, omitted waiting time, residence at the 0.5 K numerical floor, time outside the histogram range, its number of temperature samples, and the change of stored grain enthalpy. A sample outside the histogram range or an exhausted sample capacity is an error, not a truncated $P(T)$. See Section 7. for the histogram algorithms. The engine also provides `grain_radiative_equilibrium`, which balances radiation absorbed from the supplied field against thermal emission, and `mc_equilibrate`, which advances the temperature and random state before any histogram or energy accounting begins.

- `mc_sed.f90`: SED builder. Wraps an OpenMP parallel loop over the size distribution; each iteration creates its own `mc_grain_t` and `rng_t`, runs MC, and accumulates the thread-local contribution to λI_λ.
- `main_mc.f90`: standalone single-grain driver (`main_mc.x`). Reads a namelist, runs one MC, writes `<prefix>_PT.dat` and, with `record_trajectory=.true.`, the trajectory files `<prefix>_Tt.dat` and `<prefix>_evt.dat`.
- `main_mc_sed.f90`: size-distribution SED driver (`main_mc_sed.x`). Reads a namelist, calls `sed_init` from the astrodust code to load the grain Q table, size distribution, and PAH cross sections, then loops via `mc_sed`. It calls `sed_init` without the optional `lam_min`, so the wavelength grid is the plain T-matrix Q-table grid (0.0912 μm to 3.98×10^4 μm) and the extreme-ultraviolet extension available to the library does not enter here.
- `main_pT_compare.f90`: single-grain comparison driver (`main_pT_compare.x`), Section 6.5.

Table 1: Selectable heat-capacity models (`comp` key).

key	enthalpy model
`gra_dl01`	DL01 `Car0` graphite (Debye-2 with $\Theta_1 = 863$ K, $\Theta_2 = 2504$ K, plus C–H modes for $n_{\rm atom} \le 5.75 \times 10^4$, i.e. $a \lesssim 50$ Å — the Draine & Li 2001 threshold). **Default for graphite and PAH.**
`gra_d85`	Single Debye-3, $\Theta_D = 420$ K, $n_{\rm atom} = 1.14 \times 10^{23}$ cm^{-3}. Original Draine & Anderson (1985) prescription.
`sil_dl01`	DL01 silicate (`Sil`).
`ad_s1_c1`	Astrodust Stage 1, C1 prefactor (pure DL01 `Sil`).
`ad_s1_c2`	Astrodust Stage 1, C2 prefactor (density-corrected).
`ad_s2`	Astrodust Stage 2: $E_{\rm Sil}(N_{\rm Sil}) + E_{\rm Car}(N_{\rm Car})$ via volume-weighted sub-volumes ($f_C^{\rm vol} = 0.193$, $P = 0.20$).

3.2 Stationary recording and transient experiments

Each MC driver first obtains a radiative-equilibrium temperature from its configured grain and radiation field. The preparation trajectory starts at this temperature and continues for a predetermined time

$$T_w = \max\left(10\tau_{\rm relax}, \frac{100}{\Gamma_{\rm evt}}\right) \tag{12}$$

when the discrete event rate $\Gamma_{\rm evt}$ is positive. For zero event rate, $T_w = 10\tau_{\rm relax}$. The second term gives 100 *expected* absorption events, not a minimum actual count; in the graphite runs of Section 4. the preparation absorbed 84–106 photons. The relaxation time is the larger of the equilibrium enthalpy divided by absorbed power and the local thermal cooling time when absorbed power is positive; only the local cooling time is used when absorbed power vanishes. This duration does not by itself prove statistical convergence of $P(T)$. The discrete absorption rate is independent of temperature, so its events form a homogeneous Poisson process. The recording starts at T_w, which need not coincide with an event; the engine draws a fresh exponential residual waiting time, consistent with Poisson memorylessness.
The temperature and random state at the end of preparation start the recorded trajectory. Its event count and $t_{\rm max}$ are recording budgets. The preparation interval contributes neither histogram residence nor elapsed time, event absorption, emission, or stored-energy change to the reported run. The comparison driver copies this one prepared random state before calling the fixed-grid, two-pass, and buffered engines, so each begins with the same temperature and photon sequence. The standalone driver defaults to this stationary procedure; an explicit positive `T_init` instead requests a transient experiment and skips preparation. A zero `T_init`, an unknown engine name, or an unreadable or malformed namelist terminates with an error.

3.3 Energy bookkeeping

For each MC run we accumulate the event absorption $E_{\rm abs}^{\rm evt} = \sum_{\rm events} hc/\lambda$, the continuous absorption $4\pi a^2 H_{\rm cont} t$, and the emitted energy $E_{\rm emit} = \int 4\pi a^2 \langle Q_{\rm abs}\rangle_T \sigma_{\rm SB} T^4\, dt$ over the recorded trajectory only. The engines also report the change of grain enthalpy over the recorded trajectory, $\Delta E_{\rm grain}$. The signed closure fraction

$$\epsilon_E = \frac{E_{\rm abs}^{\rm evt} + 4\pi a^2 H_{\rm cont} t - E_{\rm emit} - \Delta E_{\rm grain}}{E_{\rm abs}^{\rm evt} + 4\pi a^2 H_{\rm cont} t} \tag{13}$$

separates stored energy from numerical error. It vanishes for an exact integration of Eq. (7).

Size-distribution runs. In the HD23 astrodust+PAH runs at $U = 1.585$ (Section 5.), every one of the 1,946 simulated grains of the two branch builds (default, all-MC, and 5×10^5-event runs) has zero omitted and zero outside-grid time, and ϵ_E lies between −1.084% and +0.076%. The stored term is small: $|\Delta E_{\rm grain}|$ is at most 1.2×10^{-4} of the absorbed energy. The most negative values belong to the stochastic grains: −0.73% (astrodust, $a = 18.8$ Å), −0.79% (neutral PAH, 3.5 Å), and −1.08% (cation PAH, 13.3 Å); the equilibrium-gate grains run in all-MC mode reach +0.08%. The two smallest astrodust grains ($a = 3.548$ and 3.758 Å) have $\epsilon_E = -0.700\%$ and −0.667%. The same grains give the same ϵ_E to within 3.2×10^{-6} with 5×10^5 events as with 5×10^4, so the residual is not sampling noise; it is consistent with the linearization error of the 5% cooling step (Eq. 11). Negative ϵ_E means the trajectory emits more than it absorbs.

Graphite runs. The 100 Å graphite grain at $U = 1$ with 5×10^5 recorded events (Figure 2) gives $E_{\rm emit}/(E_{\rm abs}^{\rm evt} + 4\pi a^2 H_{\rm cont} t) = 1.0040$ and $\epsilon_E = -0.40\%$; across the five sizes of that figure ϵ_E ranges from −0.29% (200 Å) to −0.54% (10 Å). The ten panels of Figure 1 are listed in Table 2.

Spectral normalization. The SED builder normalizes each stochastic grain spectrum to its absorbed power, including continuous heating, after histogram quadrature (Eq. 15). This normalization enforces the stationary energy balance of the emitted spectrum; it does not validate the sampled $P(T)$, and the closure fractions above are measured before it.

4. Validation against PIIM Figures 24.5 and 24.6

4.1 Figure 24.5: $T(t)$ trajectories

We run the standalone `main_mc.x` (script `scripts/run_fig24_5.sh`) with composition `gra_dl01` for the ten panels of Draine (2011) Figure 24.5: grain radii $\{10, 20, 50, 100, 200\}$Å crossed with $U \in \{1, 10^2\}$, the fixed cutoff $\lambda_c = 1000\,\mu$m, and trajectory recording, which uses the fixed-grid engine. Each recorded interval is $t_{\max} = 10^5$ s ($\sim$1 day), following PIIM Fig. 24.5, and starts after the stationary preparation, so $t = 0$ is a random point of the stationary history rather than a cold start.

Figure 1 reproduces the qualitative behavior of PIIM Fig. 24.5: large grains ($a \geq 100$Å) stay close to a steady temperature; smaller grains spike to hundreds of K between sparse photon events. Table 2 lists the number of photon absorptions in each recorded interval and the closure fraction of Eq. (13). The 10 Å, $U = 1$ panel absorbs no photon in 10^5 s (mean waiting time 5.1×10^5 s); its only absorbed energy is the continuous heating, so its closure fraction compares a small cooling residual with a very small denominator.

Table 2: Recorded photon absorptions $N_{\rm evt}$ and closure fraction ϵ_E (Eq. 13) of the Figure 24.5 panels, $t_{\max} = 10^5$ s.

	$U = 1$		$U = 10^2$	
a (Å)	$N_{\rm evt}$	ϵ_E (%)	$N_{\rm evt}$	ϵ_E (%)
200	1676	–0.29	171223	–0.01
100	202	–0.40	20576	–0.14
50	27	–0.35	2449	–0.26
20	1	–0.28	136	–0.32
10	0	–1.72	18	–0.45

4.2 Figure 24.6: $dP/d\ln T$

For the $P(T)$ reproduction (script `scripts/run_fig24_6.sh`) we use the same five sizes at $U = 1$ with $N_{\rm evt} = 5 \times 10^5$ recorded events per grain and the two-pass engine. The recorded intervals range from 2.9×10^7 s (200 Å) to 2.5×10^{11} s (10 Å). Figure 2 shows the time-weighted $dP/d\ln T$ distributions, reproducing the main features of PIIM Fig. 24.6: large grains have a narrow peak near $T_{\rm eq}$; small grains have broad distributions extending to several hundred K. All curves use the photon distribution described in the next subsection.

4.3 Photon wavelength distribution

The event rate and photon-wavelength distribution use the same nonnegative absorption integrand. Its cumulative probability is tabulated on the physical wavelength knots, including the $0.0912\,\mu$m Lyman edge of the Mathis field. Within each interval, the code integrates the linear photon density and inverts that integral for each absorption event. Intervals with zero probability have no sampled wavelengths. The stochastic/continuous split at λ_c is applied to this integral, so the sampled photons and the event rate have the same spectral support.

For each of the five graphite sizes, direct draws of 2×10^6 photons gave no wavelength below the Lyman edge. Their mean energies agree with the analytic mean of the implemented piecewise-linear probability distribution to within 0.60 standard errors (Table 3). The corresponding normalized cumulative integrals are unity to within 4.4×10^{-15}. The temperature limits in the table are direct extrema of the recorded trajectories; the one- and two-photon temperatures use 13.6 eV absorptions starting from each measured minimum.

Table 3: Direct photon sampling for graphite at $U = 1$ ($\lambda_c = 1000\,\mu$m), and the temperature reached by one and two 13.6 eV photons from the lowest recorded temperature $T_{\min}$ of Figure 2. The last column gives the actual maximum recorded temperature, rather than a histogram-bin edge. The sampled/exact mean compares 2×10^6 draws with the analytic mean of the implemented photon distribution.

a (Å)	unsupported draws $\lambda < 0.0912\,\mu$m	$\langle E \rangle$ ratio sampled/exact	$T_{\min}$ (K)	$T_{1\gamma}$ (K)	$T_{2\gamma}$ (K)	$T_{\max}^{\mathrm{MC}}$ (K)
10	0	0.99961	2.376	422.0	587.1	420.74
20	0	0.99961	2.390	179.6	234.2	181.45
50	0	0.99960	4.311	69.63	87.78	83.92
100	0	0.99959	9.704	35.06	44.01	45.51
200	0	0.99960	16.294	21.25	24.59	30.17

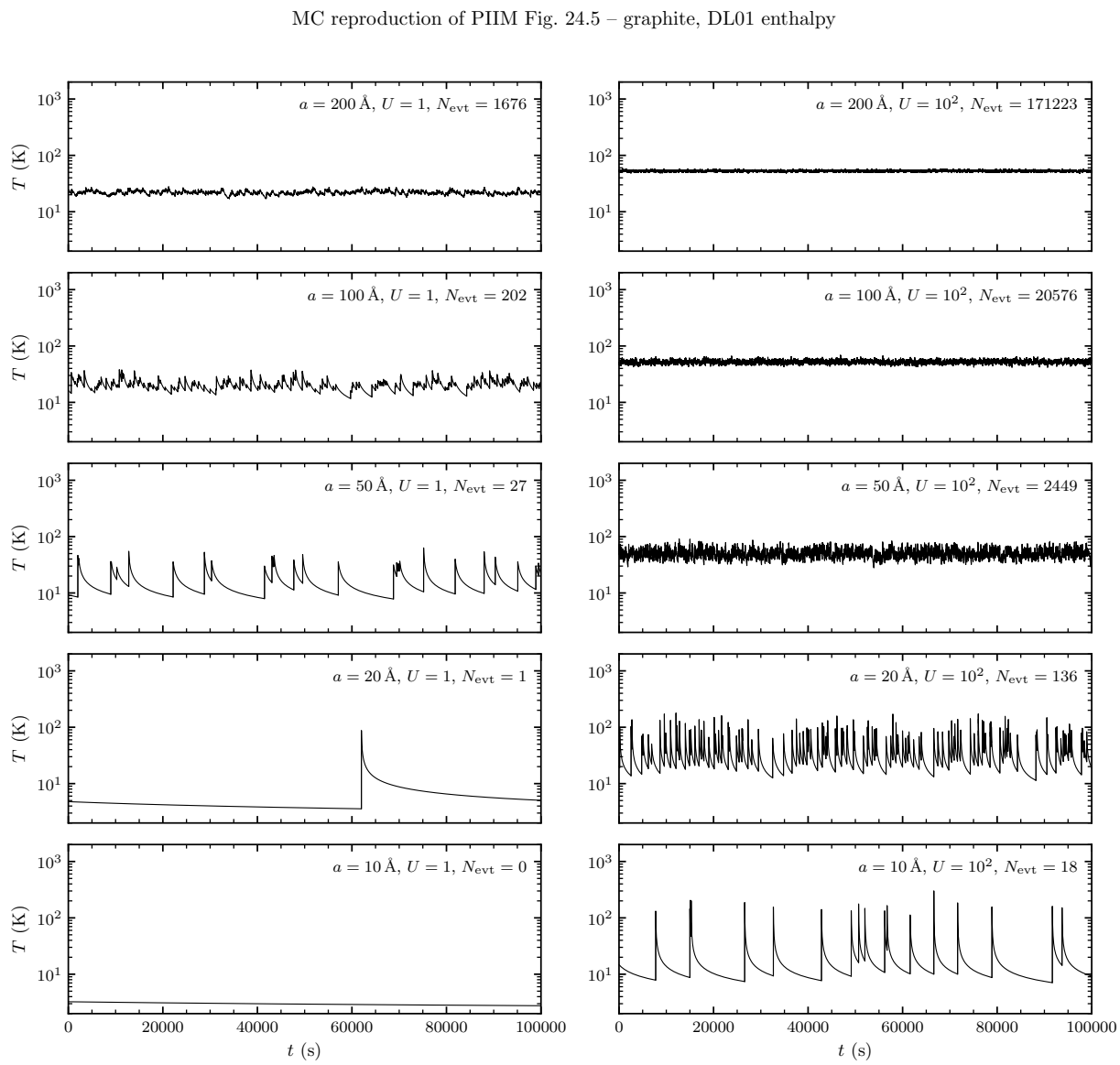


Figure 1: Reproduction of Draine (2011) Figure 24.5: MC temperature histories for graphite grains in the Mathis ISRF, recorded for 10^5 s after the stationary preparation. Black curves are the integrator sub-step samples dumped from `cool_segment` (at least 0.5 s apart; no closed-form reconstruction); vertical line segments show the photon jumps from $T_{\rm pre}$ to $T_{\rm post}$ at each absorption event. $N_{\rm evt}$ counts the absorptions in the recorded interval. Input: `mc/output/fig24_5/*_Tt.dat` and `*_evt.dat`.

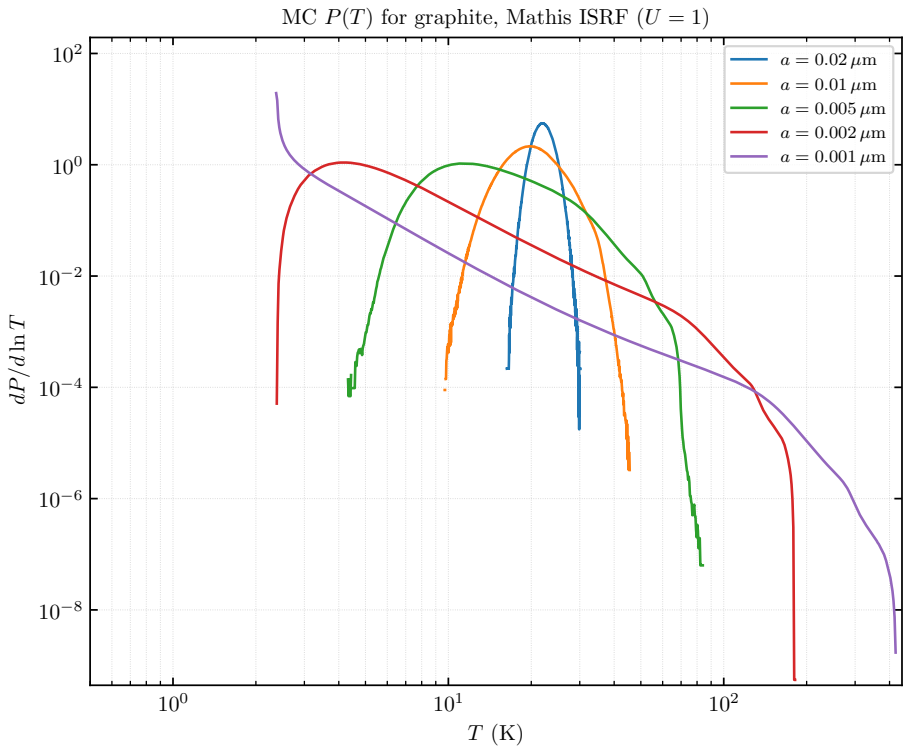


Figure 2: $dP/d\ln T$ for graphite grains in the Mathis ISRF ($U = 1$), for sizes $a \in \{0.001, 0.002, 0.005, 0.01, 0.02\}\,\mu$m, 5×10^5 recorded events each. Compare with PIIM Figure 24.6. The full occupied temperature range is shown; zero-probability bins are omitted without connecting the curves across them. Several absorptions before the grain cools to its baseline can raise it above the temperature reached by one 13.6 eV photon (Table 3). Input: `mc/output/fig24_6/*_U1_PT.dat`.

5. SED builder over a grain size distribution

The `mc_sed` module integrates $P(T)$ across the HD23 astrodust+PAH size distribution (evaluated by `sed_astrodust_mod::sed_init` from the HD23 analytic forms in `size_dist.f90`) and produces the emission SED $\lambda I_\lambda / N_\mathrm{H}$ in the same units as the production `sed_solve` routine. The size grid has 167 radii from 3.548 Å to 5.012 μm for each population.

5.1 SED accumulation

For each grain a_j in the size distribution:

$$J_\lambda \mathrel{+}= (dn/N_\mathrm{H})_j \cdot C_\mathrm{abs}(\lambda, a_j) \cdot \sum_i \frac{w_i}{t_\mathrm{total}} \cdot \pi B_\lambda(T_{\mathrm{mid},i}), \tag{14}$$

where w_i is the time spent in histogram bin i (with center $T_{\mathrm{mid},i}$) accumulated by the MC. Summing over both grain populations (astrodust + PAH) and applying the same final unit conversion as the production solver, $\lambda I_\lambda = \lambda J_\lambda \times 10^{-3}$, gives the final SED in erg s^{-1} sr^{-1} H^{-1}.

5.2 Energy normalization and charge-resolved PAH

Single-grain energy normalization. Equation (14) evaluates the Planck function at each histogram-bin midpoint. This quadrature introduces a spectral error whose sign depends on wavelength and the temperature distribution; convexity does not establish a universal overestimate. The code normalizes each stochastic grain spectrum to its absorbed power using

$$s_\mathrm{ec} = \frac{P_\mathrm{abs}}{P_\mathrm{emit}^\mathrm{gray}}, \qquad P_\mathrm{emit}^\mathrm{gray} = \sum_i w_i 4\pi a^2 \langle Q_\mathrm{abs} \rangle_T [T_{\mathrm{mid},i}] \sigma_\mathrm{SB} T_{\mathrm{mid},i}^4, \qquad P_\mathrm{abs} = \frac{E_\mathrm{abs}^\mathrm{evt}}{t_\mathrm{total}} + 4\pi a^2 H_\mathrm{cont}, \tag{15}$$

where P_abs is the grain's total absorbed power (discrete photon events plus the continuous long-wavelength heating H_cont) and $P_\mathrm{emit}^\mathrm{gray}$ is the gray bolometric emission of the same histogram (with w_i normalized to unit sum). The normalized single-grain spectrum $s_\mathrm{ec} J_\lambda^{(j)}$ emits exactly the absorbed power while keeping the spectral shape set by $P(T)$. For the two smallest astrodust grains, the histogram's gray emission before normalization is 1.0070 and 1.0066 times the absorbed power. In the default run the integrated MC PAH emission is 0.9989 times that of the production solver over the full wavelength grid.

Charge-resolved PAH populations. The production solver treats PAHs as two charge states, neutral and cation, mixed by an ionization fraction $f_\mathrm{ion}(a)$; the two have markedly different cross sections in the near-IR (the cation carries a Mattioda et al. (2005) NIR continuum that the neutral lacks). The matrix solver loops over both charge states — each with its own C_abs, Planck-mean cooling opacity, and size-distribution weight dn — and sums the emission. A grain of one charge state absorbs and emits with that state's cross section; a single blended cross section would describe a grain that does not exist. The MC builder does the same as the matrix solver: `mc_sed_solve_pah` runs the engine twice per size, once with the neutral ($C_\mathrm{abs}^\mathrm{neu}, dn^\mathrm{neu}$) and once with the cation ($C_\mathrm{abs}^\mathrm{ion}, dn^\mathrm{ion}$) at an RNG seed offset by 5000, and sums. In the default run the MC PAH emission in 1–5 µm is 0.961 times the production result (Section 6.).

5.3 OpenMP parallelism

The grain loop in `mc_sed::mc_sed_loop` uses an OpenMP parallel region with a dynamically scheduled size loop. Each thread holds private `mc_grain_t` and `rng_t` objects and accumulates a local `Jout_local(:)`. The spectral arrays are combined in a critical section; scalar diagnostics use atomic updates. With more than one thread the result is not bit-reproducible, because the order of the floating-point sums depends on thread timing.

Each population loop calls the engine once per grain that fails the equilibrium gate. In the default mode this is 78 of the 167 astrodust radii and 79 of the 167 radii of each PAH charge state (236 MC trajectories, with 265 grains on the `calc_Teq` path); in the all-MC mode it is all 501. With 16 OpenMP threads on the machine described above, the current calculations took 158–198 s in default mode, 154–213 s in all-MC mode, and 1847–2046 s for the 5×10^5-event default mode. These wall times include concurrent runs and are not a scaling measurement.

6. Comparison with the Guhathakurta–Draine matrix solver

We compare `main_mc_sed.x` with the production astrodust solver `calc_sed.x astrodust`, whose default stochastic solver (`heuristic`) is the Guhathakurta & Draine (1989) matrix solution with look-ahead narrowing of the temperature window. Table 4 lists the inputs of the two calculations item by item; they differ only in the temperature grid used for $T_{\rm eq}$ and the equilibrium gate, and in the stochastic solver itself.

Table 4: Inputs of the MC SED (`mc/output/sed_*_S2_U1.585_irem_mc.dat`) and of the reference (`sed/output/sed_astrodust_S2.dat` plus `sed_astrodust_PAH.dat`).

item	`main_mc_sed.x`	`calc_sed.x astrodust`
optics	`q_astrodust_P0.20_Fe0.00_1.400.dat` (row selection at the Lyman limit of the astrodust table)	`sedust_astrodust.h5`, cut at the Lyman limit
wavelength grid	1129 points, 0.0912–3.98×10^4 μm	same 1129 points
size distribution	HD23 eqs. (17), (24), 167 radii	same
radiation field	`J_Mathis`, corrected, $U = 1.585$, 2.725 K CMB	same
enthalpy	Stage 2	Stage 2
PAH	neutral and cation solved separately, $f_{\rm ion}(a)$	same
T grid ($T_{\rm eq}$, gate)	250 log bins, 1–3000 K	200 log bins, 2.7–5000 K
equilibrium gate	$E_{\rm grain}(T_{\rm eq}) \geq 150$ eV	same
stochastic grains	MC, two-pass, 5×10^4 events, adaptive λ_c, Eq. (15)	GD matrix, `heuristic` narrowing

6.1 Two solver modes

The `main_mc_sed.x` driver exposes a runtime namelist flag, `use_mc_for_large_grain`. Small (stochastic) grains *always* use the Monte Carlo engine; the flag selects how large (equilibrium) grains are handled:

- `use_mc_for_large_grain = .false.` (default). Each grain is first tested with the equilibrium gate $E_{\rm grain}(T_{\rm eq}) \geq 150$ eV (same threshold as `sed_solve`). Grains passing the gate are emitted as a delta-function at $T_{\rm eq}$ via `calc_Teq`. Grains failing the gate are simulated with MC.
- `use_mc_for_large_grain = .true.` All grains use MC, regardless of size. This exercises the MC engine on grains where the analytic solver applies and checks the equilibrium gate.

Table 5: Band-integrated MC/reference ratios, $\int \lambda I_\lambda \, d\ln\lambda$ over each band on the common wavelength grid (trapezoid rule).

mode	component	1–5 μm	5–30 μm	30–300 μm	300–3000 μm	all
default	total	0.961	0.998	1.001	1.001	0.999
default	astrodust	0.974	0.991	1.000	1.001	1.000
default	PAH	0.961	0.998	1.007	1.008	0.999
all-MC	total	0.961	0.998	1.001	1.001	0.999
5×10^5 events, default	total	0.960	0.998	1.001	1.001	0.999

Figure 3 and Table 5 show both modes against the reference. The band-integrated power agrees to within 0.2% in each band from 5 to 3000 μm and is 3.9% low in 1–5 μm, where the PAH emission dominates. Point by point, the total ratio stays between 0.977 and 1.011 from 5 to 3000 μm and between 0.903 and 0.977 in 1–5 μm. Below 1 μm, where λI_λ is already 2.8 decades below its peak at 7.7 μm, the ratio falls further. The all-MC mode, in which all 501 grains use MC, reproduces the default mode to 0.0003 in every band ratio. A tenfold increase of the event count (`sed_hq` run, 5×10^5 events per grain) changes the band ratios by at most 0.0014, so the 1–5 μm difference is not MC sampling noise.

6.2 Equilibrium grains in the all-MC mode

In the all-MC mode, `mc_sed_loop` writes the radiation-equivalent temperature $T_{\rm rad}^{\rm MC} = \langle T^4 \rangle_P^{1/4}$ of every grain that passes the equilibrium gate, together with its $T_{\rm eq}$, to `output/diag_Teq_vs_Trad.dat`. For the 265 such grains ($a = 0.0316$–$5.01\,\mu$m; 89 astrodust and 88 of each PAH charge state) the ratio $T_{\rm rad}^{\rm MC}/T_{\rm eq}$ lies between 0.99871 and 1.00008; above 0.1 μm it lies between 0.99989 and 1.00000. The file identifies the astrodust, neutral PAH, and ionized PAH populations in its sixth column; its rows appear in thread completion order.

6.3 When to use each mode

Default (`F`). The SED builder applies the equilibrium gate and uses `calc_Teq` for grains that pass it. Smaller grains use the selected MC engine. This is the default calculation and avoids an MC trajectory for 265 of the 501 grains.

All-MC (`T`). Every grain uses MC, including grains normally treated at $T_{\rm eq}$. This mode checks the equilibrium gate and the cooling treatment; it gives the band ratios of the default mode to 0.0003.

6.4 Step budget

The engines integrate each waiting interval in chunks of 300 steps (adaptive engines) within a budget of 10^6 steps per interval. In the default, all-MC, and 5×10^5-event SED runs, none of the 2.36×10^7, 5.01×10^7, and 2.36×10^8 cooling segments (both passes counted) needed a second chunk, and no interval was left incomplete.

6.5 Single-grain $P(T)$ cross-check

As a direct single-grain check of the MC temperature distribution against the matrix solver — independent of the size-distribution integral — the driver `main_pT_compare.x` calls, for a list of up to eight grain radii, all three MC engines (`mc_run_engine`, `mc_run_engine_2pass`, `mc_run_engine_buffered`) *and* `calc_P` on the same input ($C_{\rm abs}$, J_λ, grain enthalpy, Planck-mean absorption, and T grid) and writes four $dP/d\ln T$ blocks side by side. Its T grid has 250 logarithmic bins from 1 to 3000 K, and `calc_P` is applied to the full grid without the window narrowing of `sed_solve`. For `pop='pah'` it uses the $f_{\rm ion}$-blended carbonaceous cross section, a self-consistent single grain for comparing two solvers but not one of the two charge states of the SED. The driver prepares each grain once, saves the resulting temperature and random state, then runs each engine from that identical state. It measures the two-pass sub-sample count and allocates exactly that capacity for the buffered replay; it rejects more than 10^9 samples (approximately 16 GB for two real arrays), a replay count mismatch, and inadequate capacity rather than accepting a truncated $P(T)$. In every run reported here the three engines saw the same trajectory: the two-pass and buffered histograms are identical, and the fixed-grid histogram differs only in its bins.

Table 6: Single-grain comparisons, Mathis ISRF at $\log U = 0.20$: the maximum difference between the two-pass MC and GD cumulative distributions, and the ratio of their $\langle T^4 \rangle^{1/4}$. Wall times are from the v1.00 two-pass calculations with concurrent jobs and are not scaling measurements.

population	a (Å)	$N_{\rm evt}$	max $\lvert\Delta$CDF$\rvert$	$T_{\rm rad}$ MC/GD	MC wall (s)
astrodust	5	5×10^5	0.568	1.004	514
astrodust	10	5×10^5	0.029	1.008	286
astrodust	20	5×10^5	0.048	1.010	211
astrodust	50	5×10^5	0.030	1.007	41.8
PAH (blended)	5	5×10^5	0.030	1.005	258
PAH (blended)	10	5×10^5	0.029	1.010	160
PAH (blended)	30	5×10^5	0.028	1.007	43.4
PAH (blended)	100	5×10^5	0.039	1.004	6.7
astrodust	100	2×10^5	0.039	1.004	4.4

Figures 4 and 5 show the adaptive two-pass block for four stochastic sizes of each population, and Table 6 summarizes

the agreement. The GD cumulative distribution is evaluated with bin edges at the geometric midpoints of its T nodes. Apart from the coldest astrodust grain, the two cumulative distributions differ by at most 0.048, and the MC radiation-equivalent temperature is 0.4–1.0% higher than that of GD. For $a = 5$ Å astrodust the difference is at the cold end: the recorded MC histogram assigns 0.564 of its time at or below 1 K, mostly at the 0.5 K numerical floor, while the GD grid starts at 1 K and places 0.596 in that first node. The quoted MC wall times are for the two-pass engine on one thread (both passes).

For equilibrium grains the 250-bin grid is too coarse for the narrow physical $P(T)$: adjacent nodes differ by a factor 1.033. Applied directly, `calc_P` spreads the $P(T)$ of the grains from 501 Å to 5 μm over 14–15 nodes holding more than 10^{-3} each, about $\pm 20\%$ around $T_{\rm eq}$ (Figure 6), while the MC trajectories have standard deviations of 0.030 K (501 Å), 0.007 K (1000 Å), and below 0.001 K for $a \geq 0.3\,\mu$m. `sed_solve` resolves such grains with `calc_Teq` or by narrowing the window (`narrow_iterative` and the look-ahead narrowing in `sed_astrodust.f90`), and the default MC mode uses `calc_Teq` for them. In an all-MC run every simulated grain begins preparation at its radiative-equilibrium temperature, regardless of size. The adaptive MC engines resolve a narrow $P(T)$ when the trajectory spans more than a few bin widths; see Section 7..

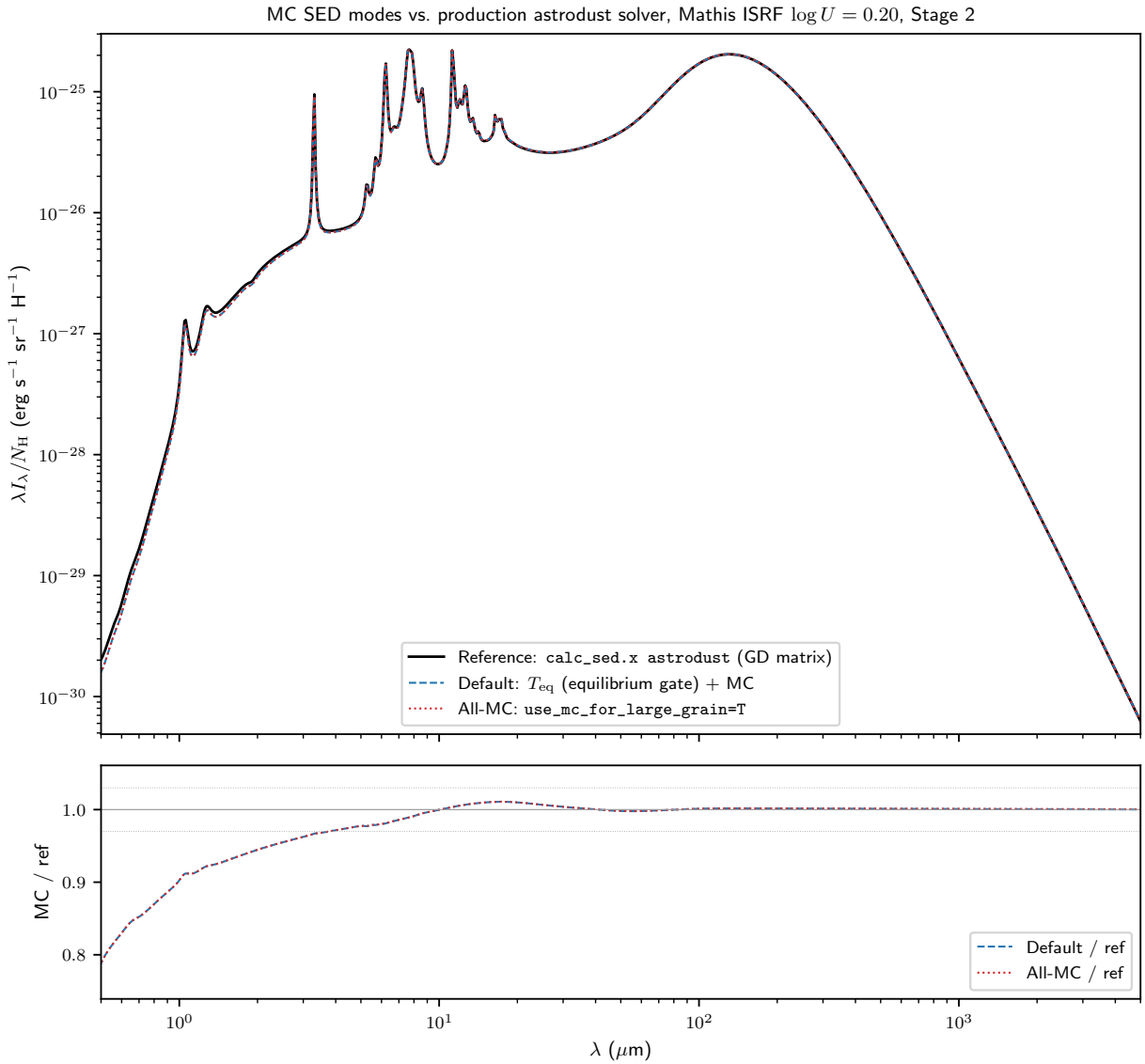


Figure 3: MC SED modes versus the production astrodust solver, for the HD23 astrodust+PAH model in the Mathis ISRF at $\log U = 0.20$, Stage 2 enthalpy, inputs of Table 4. Top: $\lambda I_\lambda / N_{\rm H}$. Bottom: ratio of each mode to the reference on the common wavelength grid (limits include all finite displayed ratios; dotted lines at $\pm 3\%$). The default (blue dashed) and all-MC (red dotted) curves overlap. Inputs: `mc/output/sed_default_S2_U1.585_irem_mc.dat`, `mc/output/sed_mc_all_S2_U1.585_irem_mc.dat`, `sed/output/sed_astrodust_S2.dat`, `sed/output/sed_astrodust_PAH.dat`.

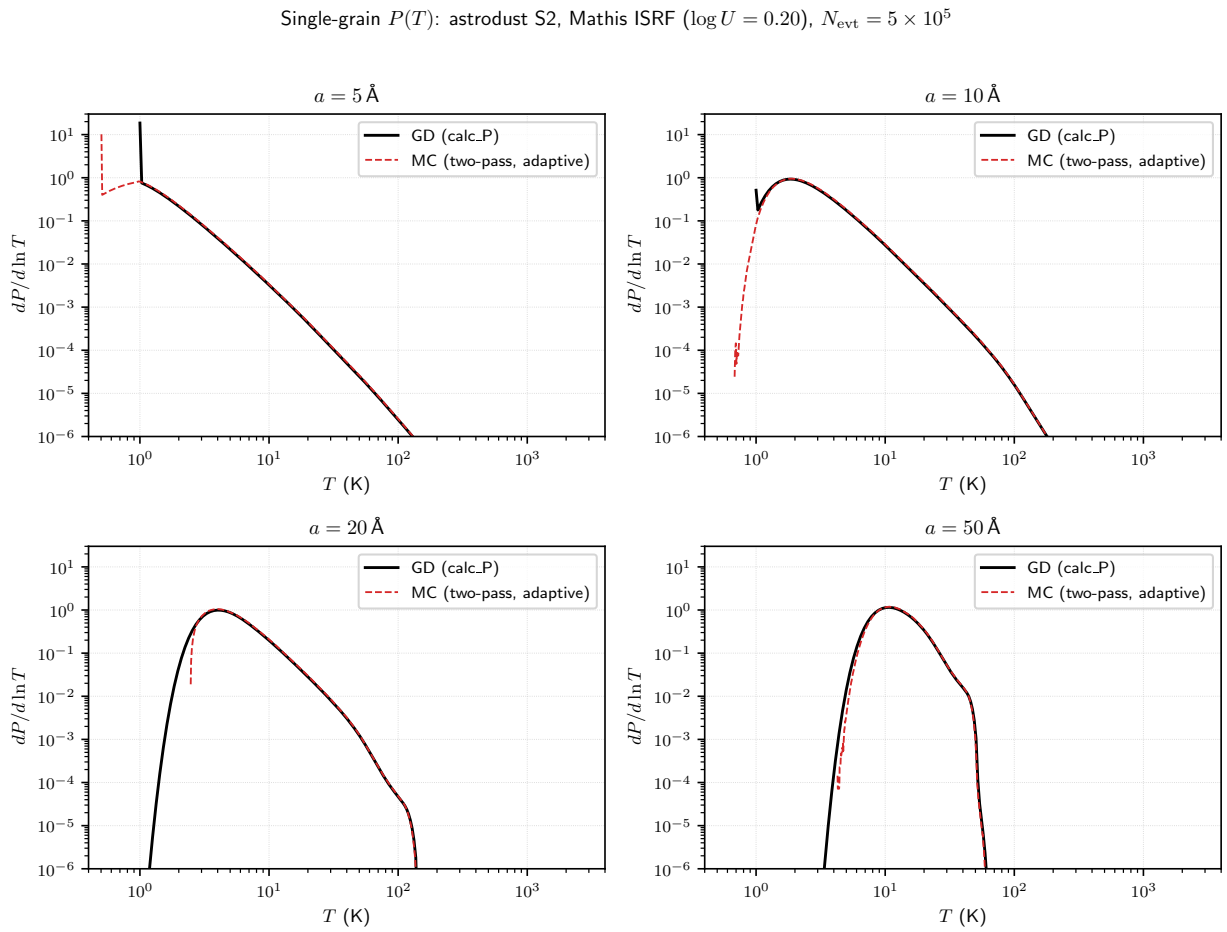


Figure 4: Single-grain $dP/d\ln T$ for the astrodust population (Stage 2 enthalpy), four radii in the stochastic regime ($a = 5, 10, 20, 50$ Å). Black: Guhathakurta–Draine matrix solver `calc_P` on the 1–3000 K grid. Red dashed: Monte Carlo (`mc_run_engine_2pass`, adaptive histogram grid, $N_{\rm evt} = 5 \times 10^5$). The spikes at the left edge are the MC residence at the 0.5 K floor and the GD mass in its 1 K node. Input: `mc/output/pT_compare_ad_0[1-4]_a*A.dat`.

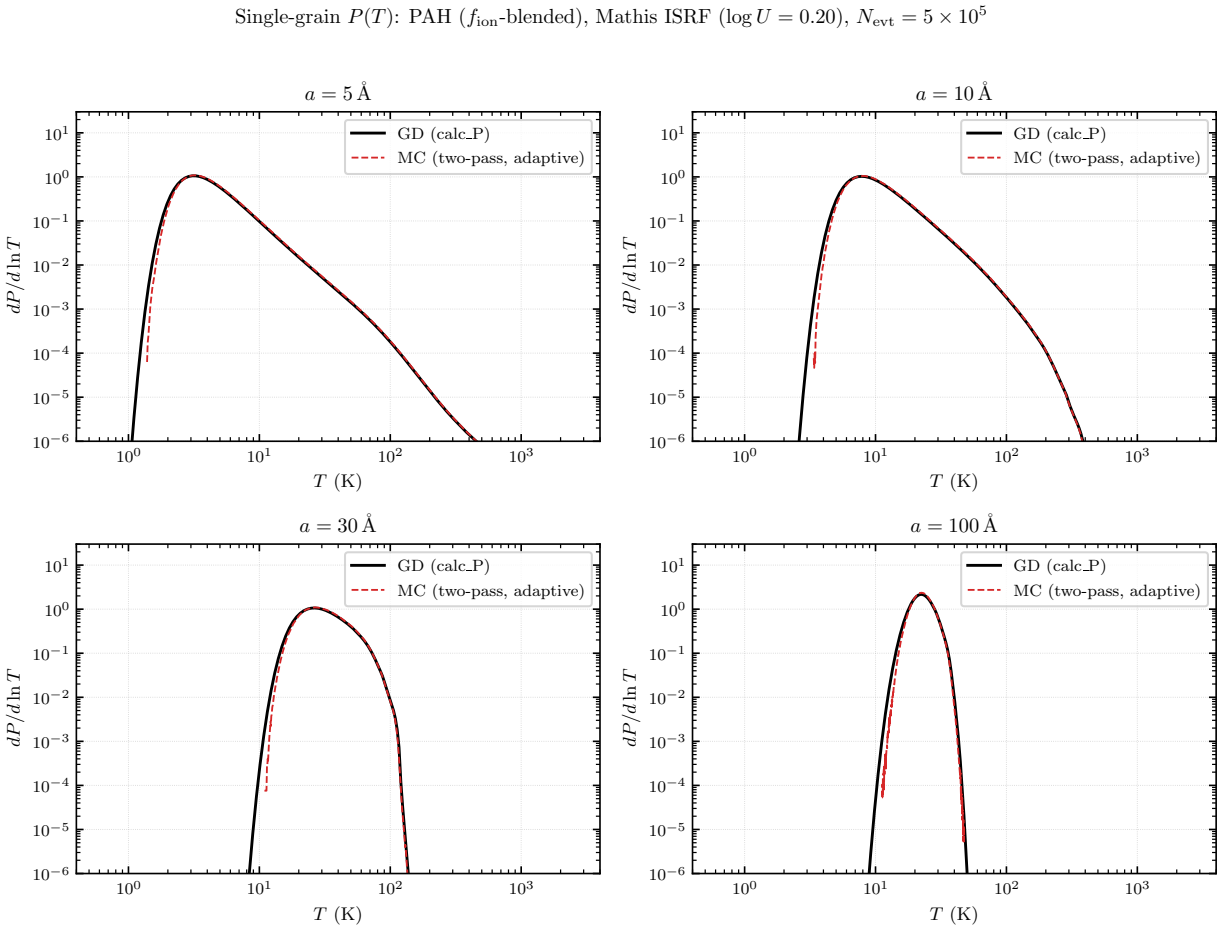


Figure 5: Single-grain $dP/d\ln T$ for the $f_{\rm ion}$-blended PAH grain, four radii ($a = 5, 10, 30, 100$ Å). Same conventions as Fig. 4 ($N_{\rm evt} = 5 \times 10^5$). At $a = 5$ Å the distribution extends to 1260 K. Input: `pT_compare_pah_0[1-4]_a*A.dat` from `run_pT_compare_pah.nml`, retained under `SEDust_JKAS/results/20261031_mc_photon_support/report_extra`.

7. Adaptive-grid MC engines

The engine `mc_run_engine` uses a fixed logarithmic histogram on $[T_{\rm min}, T_{\rm max}] = [0.5, 5000]$ K with $N_{\rm hist}$ bins set by the caller: 600 in `main_mc.x` and `main_pT_compare.x` (adjacent edges differ by a factor 1.01547) and 2000 in the SED builder (factor 1.00462). This grid suits the broad distributions of stochastically heated grains, but a narrow equilibrium distribution falls into few of its bins. In the large-grain comparison of Figure 6 (600 bins), the fixed grid places more than 10^{-3} of the time in 2 bins at 501 Å and in a single bin for every grain from 1000 Å to 5 μm; the mean temperature of such a histogram is the bin midpoint, up to 0.7% from $T_{\rm eq}$ (5 μm).
To address this, two adaptive-grid engines build the histogram grid from the observed trajectory itself.

Method A: two-pass with deterministic RNG replay (`mc_run_engine_2pass`). Pass 1 runs the full selected photon sequence without binning and records the minimum and maximum of all temperature sub-samples. The random state is then restored; Pass 2 repeats the same photon sequence, integrates each waiting interval completely, and bins every sub-sample on a grid with a margin around the measured range. Because the integrator and the photon sampler are deterministic in their inputs, Pass 2 traces the same trajectory as Pass 1, at about twice the cost of a single pass: the two-pass engine took 1.4–2.1 times the fixed-grid wall time in the concurrently run comparisons of Section 6.5. By default, $N_{\rm pass1} = N_{\rm evt}$. A shorter first pass is allowed, but any Pass 2 sample outside its grid returns an explicit range-error status; it is not silently omitted. The engine reports its complete sub-sample count for a buffered calculation with the same grain and random seed.

Method B: single-pass with sample buffer (`mc_run_engine_buffered`). The trajectory is integrated once, retaining each completed cooling step's eight $(T_{\rm sub}, dt_{\rm sub})$ pairs. Its full observed range determines the adaptive histogram grid after the run. If the caller-supplied capacity cannot hold the next complete step, the engine returns a sample-capacity status without accepting a truncated prefix as a complete $P(T)$. The single-grain comparison driver and the standalone driver use the measured sub-sample count of an identical-state two-pass run to set this capacity (for example 5.2×10^8 samples, about 8 GB, for the 5 Å astrodust grain with 5×10^5 events). For the OpenMP SED builder, capacity measurement and buffered execution are serialized so only one grain can hold the large buffer at a time. All drivers reject capacities above 10^9 samples (approximately 16 GB for two real arrays) and a count mismatch. The default SED engine is the two-pass method.

Adaptive grid spacing. Both methods call `build_adaptive_grid` with margin $\epsilon = 0.05$:

$$T_{\rm lo} = \max\left[0.5\,{\rm K}, (1-\epsilon)\,T_{\rm min}^{\rm obs}\right], \quad T_{\rm hi} = (1+\epsilon)\,T_{\rm max}^{\rm obs}. \tag{16}$$

If $T_{\rm hi}/T_{\rm lo} > 3$, the bins are log-spaced from $T_{\rm lo}$ to $T_{\rm hi}$; otherwise they are linearly spaced. The threshold separates multi-decade stochastic excursions (where log spacing resolves the high-T tail) from narrow equilibrium distributions (where linear spacing gives bins of uniform ΔT). For the grains of Fig. 6 (2×10^5 events, 600 bins) the observed ranges are [8.78, 37.29] K at 100 Å (log grid; $T_{\rm hi}/T_{\rm lo} = 4.69$ with the margins), [18.244, 18.703] K at 501 Å, and [17.551, 17.645] K at 1000 Å (linear grids); for $a \geq 0.3\,\mu$m the observed range is below 10^{-4} of T. Because the margin fixes the grid width at no less than 10% of T, a linear grid of 600 bins has bins no narrower than $1.7 \times 10^{-4}\,T$. The two-pass histogram

then places more than 10^{-3} of the time in 30 bins at 501 Å, 13 bins at 1000 Å, and 1–2 bins for $a \geq 0.3\,\mu$m, where the distribution is narrower than one bin. Its mean temperature agrees with $T_{\rm eq}$ to 0.004% for every grain from 501 Å to 5 μm. The bin lookup is $O(1)$ in both cases.

Runtime selection. `main_mc_sed.x` exposes a namelist key `mc_engine` $\in$ {`'fixed'`, `'2pass'`, `'buffered'`} that dispatches grain by grain inside the OpenMP loop. The default is `'2pass'` (adaptive). When `use_mc_for_large_grain=.true.` and `'fixed'` are requested together, the driver promotes the selection to `'2pass'` and logs a warning, because the fixed grid leaves the equilibrium grains in one or two bins. A buffered calculation must hold all required sub-samples to return a complete histogram.

Pass 1 vs Pass 2 length in `mc_run_engine_2pass`. The first pass measures the range of the selected trajectory and normally covers all $N_{\rm evt}$ events. A shorter probe is accepted only if every replay sample still falls on its adaptive grid; otherwise the engine returns a histogram-range error.

7.1 Adaptive DA85 cutoff $\lambda_c(a)$

The Draine & Anderson (1985) stochastic/continuous split (§2.) assigns photons below λ_c to discrete absorption events and longer-wavelength absorption to $H_{\rm cont}$. A fixed cutoff is not appropriate for every grain size. When a single absorbed photon changes the temperature by much less than $T_{\rm eq}$, the grain responds to the mean heating rate, and resolving each photon is both unnecessary and expensive. Conversely, assigning such weak photons to rare discrete events lets the temperature relax toward the equilibrium of the remaining continuous heating between events. The SED builder and the comparison driver therefore use

$$\boxed{\lambda_c(a) \;=\; \frac{hc}{\delta T_{\rm thresh}\, T_{\rm eq}(a)\, C_{\rm tot}\big(T_{\rm eq}(a),\, a\big)},} \tag{17}$$

with $\delta T_{\rm thresh} = 0.01$. Photons with λ short enough that their single-event jump at $T_{\rm eq}$ exceeds $\delta T_{\rm thresh}\, T_{\rm eq}$ are kept stochastic; everything longer is folded into $H_{\rm cont}$. $T_{\rm eq}(a)$ is supplied by `calc_Teq` on the same single-grain cross sections; $C_{\rm tot}$ is the grain heat capacity at $T_{\rm eq}$. λ_c is clamped to the second and second-to-last points of the wavelength grid, 0.09226 and $3.936 \times 10^4\,\mu$m, so the split integrals have at least one bin on each side. This is a numerical division of the absorbed spectrum, not a physical boundary: both parts are absorbed by the grain.

At fixed temperature and material, a heat capacity proportional to a^3 makes λ_c decrease as a^{-3}; the actual cutoff also depends on $T_{\rm eq}(a)$ and the composition. For astrodust in the Mathis field at $U = 1.585$ the measured cutoff is $3.936 \times 10^4\,\mu$m (the long-wavelength clamp) at 5 Å, 7491 μm at 10 Å, 933 μm at 20 Å, 57.9 μm at 50 Å, 7.08 μm at 100 Å, and the short-wavelength clamp 0.0923 μm for every grain from 501 Å to 5 μm. For those large grains essentially all absorption is continuous heating, and the trajectory relaxes to $T_{\rm eq}$ (Section 6.).

For the two smallest astrodust grains of the size grid ($a = 3.548$ and 3.758 Å), λ_c reaches the long-wavelength clamp. Their continuous heating is $H_{\rm cont} = 5.4944 \times 10^{-19}$ and $5.8200 \times 10^{-19}\,{\rm erg\,cm^{-2}\,s^{-1}}$, which is 4.3×10^{-14} of their event power. The root of $\langle Q_{\rm abs}\rangle_T \sigma T^4 = H_{\rm cont}$ with the implemented $\langle Q_{\rm abs}\rangle_{1\,\rm K}$ is 0.02976 K for both, below both the 0.5 K floor and the 1 K lower bound of the $\langle Q_{\rm abs}\rangle_T$ table; it is not a validated physical dust temperature. In the default SED run these grains spend 0.546 and 0.473 of the recorded time at the floor, and their two-pass histograms hold 0.592 and 0.521

of the time in bins below 0.55 K. Repeating the two calculations with the same seeds and preparation (a separate driver that calls the same engine routines and reproduces the logged event power, elapsed time, and floor time of the SED run exactly), the gray emission of those cold bins is 2.1×10^{-9} and 1.8×10^{-9} of the absorbed power. The floor therefore does not change these grains' emitted power or spectrum at any measurable level, but the cold part of their $P(T)$ is a numerical construct of the floor and of the 1 K table limit, not a prediction. The same grains reach 5140 K and 4180 K after their hottest photon absorptions; the fixed-grid engine, whose grid ends at 5000 K, therefore refuses the first of them with a histogram-range status, while the adaptive engines accept both.

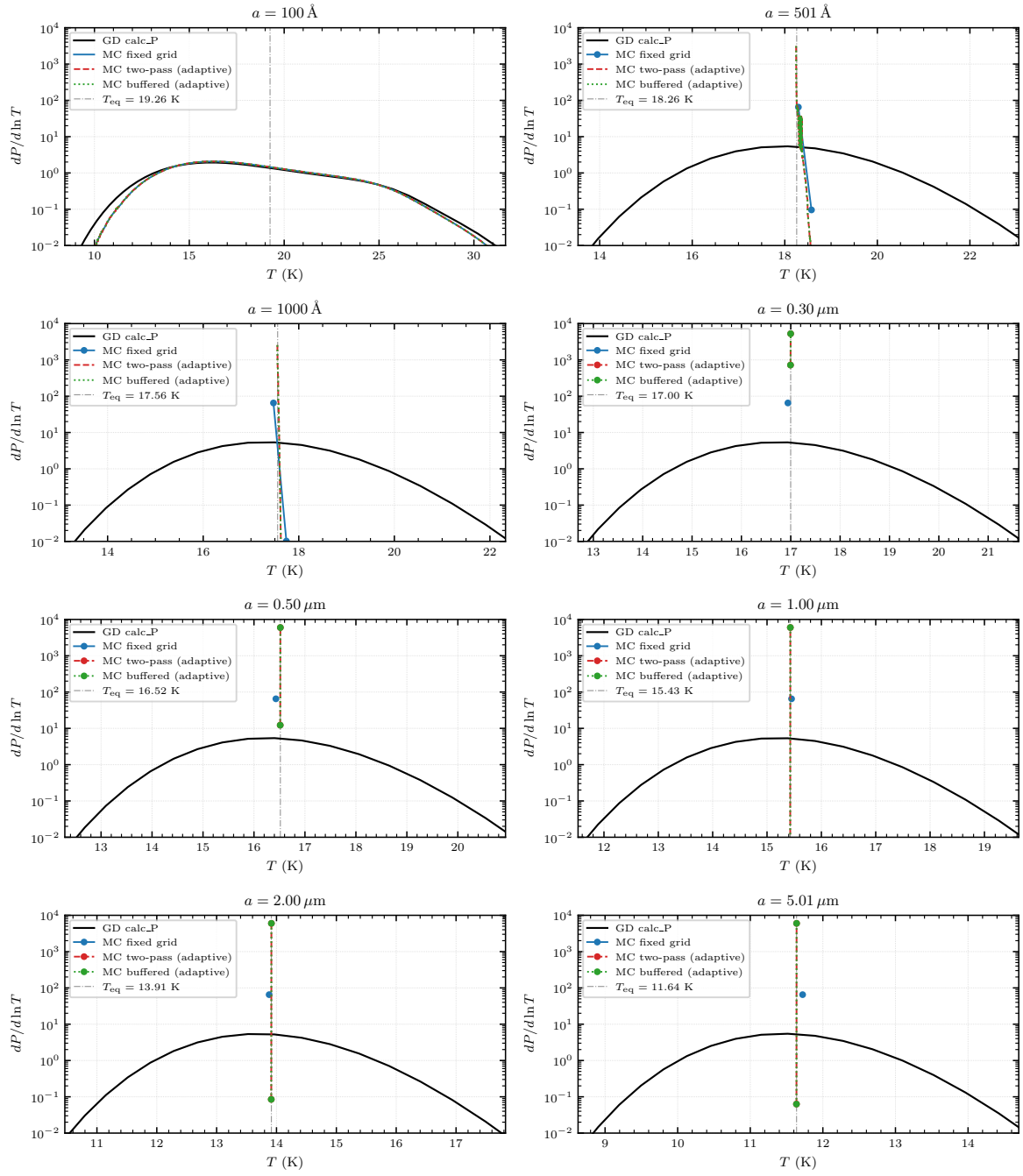


Figure 6: Single-grain $dP/d\ln T$ for large astrodust grains ($a \in \{100, 501, 1000\,\text{Å}, 0.30, 0.50, 1.00, 2.00, 5.01\,\mu\text{m}\}$, Stage 2 enthalpy, Mathis ISRF at $\log U = 0.20$, $N_{\rm evt} = 2 \times 10^5$, 600 bins). Gray dash-dotted line: $T_{\rm eq}$ from `calc_Teq`. Black: GD `calc_P` on the 1–3000 K grid; blue: fixed-grid MC; red dashed: two-pass adaptive; green dotted: buffered adaptive (identical to two-pass). Histograms with fewer than three occupied bins are drawn as points. The MC distributions are centered on $T_{\rm eq}$; the GD width reflects its grid spacing. Input: `pT_compare_large_0[1-8]_a*A.dat` from `run_pT_compare_large.nml`, retained under `SEDust_JKAS/results/20261031_mc_photon_support/report_extra`.

References

Draine, B. T., & Anderson, N. 1985, ApJ, 292, 494, “Temperature fluctuations and infrared emission from interstellar grains”.

Draine, B. T. 2011, *Physics of the Interstellar and Intergalactic Medium* (Princeton: Princeton University Press), Chap. 24.

Guhathakurta, P., & Draine, B. T. 1989, ApJ, 345, 230, “Temperature fluctuations in interstellar grains. I. Computational method”.

Draine, B. T., & Li, A. 2001, ApJ, 551, 807, “Infrared emission from interstellar dust. I. Stochastic heating of small grains”.

Mattioda, A. L., Allamandola, L. J., & Hudgins, D. M. 2005, ApJ, 629, 1188, “The near-infrared spectrum of polycyclic aromatic hydrocarbon cations”.

Draine, B. T. 2016, ApJ, 831, 109, “Graphite Revisited”.

Hensley, B. S., & Draine, B. T. 2023, ApJ, 948, 55, “The astrodust+PAH model”.